Higher order linear initial value problem (IVP): Green's function

ODE: $y'' + P(x)y' + Q(x)y = f(x)$ —— (1)

Initial condition: $y(x_0) = y_0$, $y'(x_0) = y_1$

G.S $y = y_c + y_p$

$\therefore y_c = C_1 y_1(x) + C_2 y_2(x)$

$y_p = u_1(x) y_1(x) + u_2(x) y_2(x)$ —— (2)

From the method of variation of parameter, we know

$$u_1'(x) = -\frac{y_2(x) f(x)}{W(x)}$$

$$u_2'(x) = \frac{y_1(x) f(x)}{W(x)}$$

The linear independence of $y_1(x)$ and $y_2(x)$ on the interval I guarantees that the Wronskians $W(x) = W(y_1(x), y_2(x)) \neq 0$ for all x in I. If x and x_0 are numbers in I, then integrating the

derivatives $u_1'(x)$ and $u_2'(x)$ in 3 on the interval $[x_0, x]$ and substituting into the 2, we get

$$y_p = y_1(x)\int_{x_0}^{x} \frac{-y_2(t)f(t)}{W(t)}\,dt + y_2(x)\int_{x_0}^{x} \frac{y_1(t)f(t)}{W(t)}\,dt$$

where $W(t) = \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2(t) \end{vmatrix}$

$$y_p = \int_{x_0}^{x} \frac{-y_1(x)\,y_2(t)\,f(t)}{W(t)}\,dt + \int_{x_0}^{x} \frac{y_2(x)\,y_1(t)\,f(t)}{W(t)}\,dt$$

$$= \int_{x_0}^{x} \left[\frac{-y_1(x)\,y_2(t)}{W(t)} + \frac{y_2(x)\,y_1(t)}{W(t)}\right] f(t)\,dt$$

$$= \int_{x_0}^{x} \frac{y_1(t)\,y_2(x) - y_1(x)\,y_2(t)}{W(t)}\,f(t)\,dt$$

$$= \int_{x_0}^{x} G(x,t)\,f(t)\,dt$$

where $G(x,t) = \dfrac{y_1(t)\,y_2(x) - y_1(x)\,y_2(t)}{W(t)}$

is called the Green's function

for the differential equation (1).

Ex: (1) $y'' - y = \frac{1}{x}$, (1) $y(1)=0$, $y'(1)=0$

A.E $m^2 - 1 = 0$, $m = \pm 1$

$y_c = c_1 e^{-x} + c_2 e^{x}$, Here $y_1(x) = e^{-x}$
$y_2(x) = e^{x}$

$$W = \begin{vmatrix} e^{-x} & e^{x} \\ -e^{-x} & e^{x} \end{vmatrix} = 1+1 = 2$$

The Green's function of (1) is

$$G(x,t) = \frac{y_1(t)\,y_2(x) - y_1(x)\,y_2(t)}{W}$$

$$= \frac{e^{-t}e^{x} - e^{-x}e^{t}}{2}$$

$$\therefore y_p = \int_{x_0}^{x} G(x,t)\,f(t)\,dt$$

$$= \int_{x_0}^{x} \frac{e^{-t}e^{x} - e^{-x}e^{t}}{2}\,\frac{1}{t}\,dt$$

$$= \frac{e^{x}}{2}\int_{1}^{x} \frac{e^{-t}}{t}\,dt - \frac{e^{-x}}{2}\int_{1}^{x} \frac{e^{t}}{t}\,dt$$

Ans

Ex 2: $y'' + 4y = x \quad y(0) = 0,\ y'(0) = 0$

A.E, $m^2 + 4 = 0 \Rightarrow m = \pm 2i$

$y_c = A\cos 2x + B\sin 2x$

$$W = \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix} = 2$$

Here,

$$G(x,t) = \frac{y_1(t)\,y_2(x) - y_1(x)\,y_2(t)}{W}$$

$$= \frac{\cos 2t \sin 2x - \cos 2x \sin 2t}{2}$$

$$= \frac{1}{2}\sin(2x - 2t) = \frac{1}{2}\sin 2(x-t)$$

$$\therefore\ y_p = \frac{1}{2}\int_0^x \sin 2(x-t)\, f(t)\,dt$$

$$= \frac{1}{2}\sin 2x \int_0^x t\cos 2t\,dt - \frac{1}{2}\cos 2x \int_0^x t\sin 2t\,dt$$

$$= \frac{1}{2}\sin 2x\left[-2t\sin 2t + 4\cos 2t\right]_0^x - \frac{1}{2}\cos 2x\left[t\,2\cos 2t - 4\sin 2t\right]_0^x$$

t	$\cos 2t$	$+$
1	$-2\sin 2t$	$-$
0	$-4\cos 2t$	

$-2t\sin 2t + 4\cos 2t$

t	$\sin 2t$	$+$
1	$2\cos 2t$	$-$
0	$-4\sin 2t$	

$t\,2\cos 2t - 4\sin 2t$

$\boxed{y_c = A\cos 2x + B\sin 2x}$

$$\therefore\ \boxed{y_p(x) = \frac{1}{2}x - \frac{1}{8}\sin 2x}$$

24 $y'' + y = \sec^2 x$ (1) $y(\pi) = \frac{1}{2}$, $y'(\pi) = -1$

A.E, $m^2 + 1 = 0$, $m = \pm i$

$y_c = A\cos x + B\sin x$

$W = 1$

The Green's function of (1) is

$$G(x,t) = \frac{y_1(t)y_2(x) - y_1(x)y_2(t)}{W}$$

$$= \frac{\cos t \sin x - \cos x \sin t}{1}$$

$$= \sin(x - t)$$

$$y_p = \int_{\pi}^{x} G(x,t) f(t)\, dt$$

$$= \int_{\pi}^{x} \cos t \sin x \sec^2 t\, dt - \int_{\pi}^{x} \sin t \cos x \sec^2 t\, dt$$

$\cos t = u$
$-\sin t\, dt = du$
$-\int u^{-2} du$
$-\frac{u^{-2+1}}{-2+1}$
$\cos t$

$$= \sin x \int_{\pi}^{x} \frac{\cancel{\cos t}}{\cos^{\cancel{2}} t}\, dt - \cos x \int_{\pi}^{x} \frac{\sin t}{\cos^2 t}\, dt$$

$$= \sin x \int_{\pi}^{x} \sec t\, dt - \cos x \int_{\pi}^{x} \frac{\sin t}{\cos^2 t}\, dt$$

$$= \sin x \left[\ln(\sec t + \tan t)\right]_{\pi}^{x} - \cos x \left[\frac{1}{\cos t}\right]_{\pi}^{x}$$

$$y_p = \sin x \left\{ \ln(\sec x + \tan x) - \ln(\sec \pi + \tan \pi) \right\}$$

$$- \cos x \left(\frac{1}{\cos x} - \frac{1}{\cos \pi} \right)$$

$$= \sin x \ln(\sec x + \tan x) - \sin x \cancel{\ln}^{0}(1+0)$$

$$- \cos x - \cos x$$

$$y_p = \sin x \ln(\sec x + \tan) - \cos x - 1$$

G.S $\quad \underline{y = y_c + y_p}$

(27) $x^2y'' - 2xy' + 2y = x$ — (1) $y(1) = 2,\ y'(1) = -1$

Let

$x = e^t, \quad t = \ln x$

$D(D-1)y - 2Dy + 2y = e^t$

$\Rightarrow (D^2 - D)y - 2Dy + 2y = e^t$

$\Rightarrow (D^2 - 3D + 2)y = e^t$

A.E $m^2 - 3m + 2 = 0$

$m^2 - 2m - m + 2 = 0$

$(m-2)(m-1) = 0$

$m = 1, 2$

$y_c = c_1 e^{t} + c_2 e^{2t} = c_1 x + c_2 x^2$

(1) $\Rightarrow$ $y'' - \frac{2}{x}y' + \frac{2}{x^2}y = \frac{1}{x}$

$W = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = 2x^2 - x^2 = x^2$

$y_p = \int_1^x G(x,t)\, f(t)\, dt$

$= \int_1^x \frac{y_1(x)\, y_2(t) - y_1(t)\, y_2(x)}{W(t)}\, dt$

$$y_p = x\int_1^x \frac{\cancel{t^2}\,dt}{\cancel{t^2}\cdot t} - x^2\int_1^x \frac{\cancel{t}}{t^2\cancel{t}}\,dt$$

$$= x\int_1^x \frac{dt}{t} - x^2\int_1^x \frac{1}{t^2}\,dt$$

$$= x\left[\ln t\right]_1^x - x^2\left[\frac{t^{-1}}{-1}\right]_1^x$$

$$= x\left(\ln x - \ln 1\right) + x^2\left(\frac{1}{x} - \frac{1}{1}\right)$$

$$y_p = x\ln x + x - x^2$$

$\therefore$ G.S. $y = y_c + y_p$